Unit :- 4

Algebric Structure

Binary Composition and its properties, definition of Algebric structure ; Groupoid, Semi group, Monoid, Groups, Abelian Group, properties of group.

* Binary Composition on a Set | Binary Composition

Let G be a non-empty set. Then $G \times G = \{(a,b) : a \in G, b \in G\}$

If $o : G \times G \to G$, then o is said to be a Binary operation on the set G. The image of the ordered pair (a, b) under the function o is denoted by $a o b$.

Often we use symbol $+, \times, \cdot, o, *, \oplus, \cup, \cap, \vee, \wedge$, etc. to denote Binary operation on a set.

This "+" will be a Binary operation on G iff

$a+b \in G \;\; \forall \; a, b \in G$ and $a+b$ is unique.

Similarly, "*" will be a Binary operation on G iff

$a * b \in G \;\; \forall \; a, b \in G$ and $a * b$ is unique.

A Binary Operation on a set G is sometimes also called as Binary Composition in the set G. if "*" is Binary composition in G, then $a * b \in G \;\; \forall \; a, b \in G$.

$\therefore$ G is closed w.r.t the composition denoted by "*".

* Algebraic Structure :-

A non-empty set G equipped with one or more Binary operat^n is called Algebraic Structure.

Suppose, * is Binary operation in G. Then $(G, *)$ is an Algebraic Structure.

Ex:- $(N, \neq)$, $(I, *)$, $(R, +, \cdot)$ $(I, -)$ all are Algebraic Structure

The Algebraic structure with only one Binary operation is called group.

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GROUP :-

An Algebraic structure (G, o) where G is a non-empty set. Then G is said to be group. if it satisfy the following properties where o is Binary operation;

G_1 : **Closure Property** :-

If a and b are belong to G then $a o b$ also belongs to G, ie,

$\Rightarrow a \in G \quad , b \in G$

$\Rightarrow a o b \in G \quad , \forall a, b \in G.$

G_2 : **Associative Property** :-

The Binary operation "o" is associative, if a, b, c are the elements of G then,

$$(a o b) o c = a o (b o c) \quad \forall a, b, c \in G$$

G_3 : **Existence of Identity** :-

There exist an element e in G such that

$e o a = a = a o e \quad \forall a \in G$. The element e is called Identity.

If $a o e = a \ \forall a \in G$ then e is the right identity and when $e o a = a$ then e is the left identity in G.

G_4 : **Existence of Inverse** :-

For each element $a \in G$, there exists an element of G called the Inverse of a and denoted by a^{-1} such that

$$a^{-1} o a = e = a o a^{-1}$$

where e is the identity element for "o".

{NOTE :- Identity for Binary Operation (+) is 0 (zero) and Identity for Binary operation (·) is 1 (one).

※ Abelian Group :-

A group (G, 0) is said to be Abelian; if in addition to four group properties the following property also satisfies :-

G_5: Commutative Property :-

The Binary operation "0" is commutative in G i.e,

$$a o b = b o a$$
$$\forall\, a, b \in G.$$

The Algebric structure (G, 0) satisfying properties from G_1 to G_5 is called Abelian group.

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Condition I :-

If operation is of Addition (+) :-

1) closer property :-

$a, b \in G$

$a + b \in G$

$\forall\, a, b \in G.$

2) Associative Law :-

$a, b, c \in G$

$(a+b)+c = a+(b+c)$

$\forall\, a, b, c \in G$

3) Existence of Identity :-

$a + e = a = e + a$

$\forall\, a \in G$

{$e = 0$ in Additive operation}

4) Existence of Inverse :-

$a + (-a) = e = (-a) + a$

$\forall\, a \in G.$

5) Commutative Law :-

for Abelian :-

$a + b = b + a$

$\forall\, a, b \in G.$

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Condition II :-

If operation is of Multiplication $(\times);(\cdot),(*)$:-

1) Closer property :- $a, b \in G$

$$a \cdot b \in G$$
$$\forall\ a, b \in G$$

2) Associative property :- $a, b, c \in G$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$
$$\forall\ a, b, c \in G.$$

3) Existence of Identity :- $a \cdot e = a = e \cdot a$

$$\forall\ a \in G$$

{ $e = 1$ for multiplicative operation }

4) Existence of Inverse :- $a \cdot a^{-1} = e = a^{-1} \cdot a$

$$\forall\ a \in G.$$

5) Commutative Property / Law :- $a \cdot b = b \cdot a$

$$\forall\ a, b \in G.$$

Que :- Show that the set I of all Integers (positive or negative Including zero) i.e,

$$I = \{ \text{------} -4, -3, -2, -1, 0, 1, 2, 3, 4 \text{------} \}$$

is an Infinite Abelian group w.r.t operation of addition of Integers.

Sol :- The set of given Integer is

$$I = \{ \text{----} -2, -1, 0, 1, 2 \text{----} \}$$

we have to prove that, $(I, +)$ is Abelian group.

(i) closer Property :- $a \in I$, $b \in I \Rightarrow a + b \in I\ \forall a, b \in I$

$$2 \in I,\ 3 \in I \Rightarrow 2+3$$
$$\Rightarrow 5 \in I$$
$$\forall\ 2, 3 \in I$$

(2) Associative Law :- $a \in I$, $b \in I$, $c \in I \Rightarrow$

$$(a+b)+c = a+(b+c)$$
$$\forall\, a,b,c \in I$$

$\Rightarrow$ $2 \in I$, $3 \in I$, $4 \in I$

$$(2+3)+4 = 2+(3+4)$$
$$5+4 = 2+7$$
$$9 = 9$$
$$\forall\, 2,3,4 \in I \text{ and } 9 \in I.$$

(iii) Existence of Identity :-

$$a+e = a = e+a$$
$$2+0 = 2 = 0+2$$
$$\forall\, 0,2 \in I$$

(iv) Existence of Inverse :- $a \in I \; \exists (-a) \in I$

$$a+(-a) = e = (-a)+a$$

$\therefore$ $3 \in I \; \exists (-3) \in I$

$$3+(-3) = 0 = (-3)+3$$
$$\forall\, 3, -3 \in I$$

Hence, it satisfy 4 properties, then $(I, +)$ is group.

Now, for Abelian we have to satisfy the 5^{th} property i.e,

(v) Commutative Law :- $a, b \in I$

$$a+b = b+a \quad \forall\, a,b \in I$$

$2, 3 \in I$

$$2+3 = 3+2$$
$$5 = 5 \quad \forall\, 2,3,5 \in I$$

Hence, it is Abelian group.

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Que :- Show that set of N of all natural No. $\{1, 2, 3 \ldots\}$ is not a group w.r.t addition.

Sol :- The set N of natural No. i.e

$$N = \{1, 2, 3, 4 \ldots\}$$

we have to show the $(N, +)$ is group.

(i) **Closure Property :-** $a, b \in N \Rightarrow a + b \in N$

$2, 3 \in N \Rightarrow 5 \in N \quad \forall\, 2, 3 \in N$

(ii) **Associative Law :-** $a, b, c \in N \rightarrow (a+b)+c = a+(b+c)$

$1, 2, 3 \in N \Rightarrow (1+2)+3 = 1+(2+3)$

$6 = 6 \quad \forall\, 1, 2, 3, 6 \in N$

(iii) **Existence of Identity :-**

There exist no natural No. $e \in N$ such that;

$e + a = a = a + e \quad \forall\, a \in N$

for the addition of No. the number 0 is the identity and $0 \notin N$.

$\therefore (N, +)$ is not a group.

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Example 3. *Show that the set*

$$G = \{\ldots\ldots, -4m, -3m, -2m, -m, 0, m, 2m, 3m, 4m, \ldots\}$$

of multiples of integers by a fixed integer m is a group with respect to addition.

OR

The set $G = \{km : m$ is a fixed integer and $k = 0, \pm 1, \pm 2, \ldots\}$ of multiples of integers by a fixed integer m is a group w.r.t. addition composition.

Solution. G_1. **Closure property.** Let a, b be any two elements of G. Then $a = rm$ and $b = sm$ where r and s are some integers.

Now $a + b = rm + sm = (r + s)\, m$, where $(r + s)$ is also an integer, therefore $a + b \in G$. Thus

$$a + b \in G \quad \forall\, a, b \in G.$$

Therefore G is closed with respect to addition.

G_2. **Associativity.** The elements of G are all integers and we know that the addition of integers is an associative composition.

G_3. **Existence of identity.** $0 \in G$ and we have

$$0 + a = a = a + 0 \quad \forall\, a \in G.$$

Therefore 0 is the identity element.

G_4. **Existence of inverse.** Let rm be an arbitrary element of G where $r \in I$. Then

$$rm \in G \Rightarrow (-r)\, m \in G \qquad [\because\ -r \text{ is also an integer}]$$

Also $\quad (-r)\, m + rm = (-r + r)\, m = 0m = 0$

and $\quad rm + (-r)\, m = (r - r)\, m = 0.$

$\therefore$ $(-r)\, m$ is the **additive inverse** of rm.

Thus every element of G possesses unique additive inverse.

Hence G is a group with respect to addition.

※ Some definitions :-

(i) Groupoid :-

An Algebraic Structure (G, o) is called Groupoid, Binary algebra, if the Binary operation "o" satisfies only 1 property i.e closure,

$$a, b \in G \Rightarrow a o b \in G \quad \forall a, b \in G.$$

(ii) Semi - Group :-

An Algebraic structure (G, o) is called as Semi - Group, if the Binary operation "o" satisfies 2 properties i.e,

* closure :- $a, b \in G \Rightarrow a o b \in G \quad \forall a, b \in G$

* Associative :- $a, b, c \in G.$
$\Rightarrow (a o b) o c = a o (b o c)$
$\forall a, b, c \in G.$

(iii) Monoid :-

An Algebraic Structure (G, o) is called a Monoid if the Binary operation "o" satisfies 3 properties. i.e,

* closure :- $a, b \in G \Rightarrow a o b \in G \quad \forall a, b \in G$

* Associative :- $a, b, c \in G \Rightarrow (a o b) o c = a o (b o c)$
$\forall a, b, c \in G$

* existence of Identity :- $a \in G$
$a o e = a = e o a$
$\forall a, e \in G.$

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Que 3- Show that the set of all +ve Rational No. forms an Abelian group under composition defined by

$$\left[a * b = \frac{ab}{2}\right]$$

Sol:- Given :- Set of All positive Rational No. such that

$$a*b = \frac{ab}{2} \quad \forall\ a,b \in Q_+$$

we have to prove that, $(Q_+, *)$ is Abelian group.

i) closer property :- $a \in Q_+ \ , \ b \in Q_+$

$$a*b = \frac{ab}{2} \in Q_+$$

ii) Associative Law :- $a, b, c \in Q_+$

$$(a*b)*c = \left(\frac{ab}{2}\right)*c$$

$$= \frac{\left(\frac{ab}{2}\right)\cdot c}{2}$$

$$= \frac{a\cdot\left(\frac{bc}{2}\right)}{2}$$

$$= a * \frac{bc}{2}$$

$$(a*b)*c = a*(b*c)$$

iii) Existence of Identity :- $a*e = a = e*a$

$$a*e = a$$

$$\frac{ae}{2} = a$$

$$ae = 2a$$

$$ae - 2a = 0$$

$$a(e-2) = 0$$

$$a \neq 0 \ , \ e-2 = 0$$

$$\boxed{e = 2}$$

$$a * e = \frac{ae}{2} = \frac{a \cdot \cancel{2}}{\cancel{2}} = a$$

Hence, $\boxed{a * e = a = e * a}$

(iv) Existence of Inverse :-

Let, a be any element of Q_+. If the no. b is to be the inverse of a then we have.

$$a * b = e = b * a$$

$$a * b = e$$

$$\frac{ab}{2} = 2$$

$$ab = 4$$

Inverse, $\boxed{b = \frac{4}{a}}$

$$a \in Q_+ \quad , \exists\, b \in Q_+$$

$$a * b = \frac{ab}{2}$$

$$= \frac{\cancel{a}}{\cancel{2}} \times \frac{\cancel{4}^{2}}{\cancel{a}} = 2$$

$$a * b = 2$$

$$\boxed{a * b = e}$$

Hence, $\boxed{a * b = e = b * a}$

The set of Rational No. Satisfies all 4 properties in composition $\frac{ab}{2} = a * b$

Hence, it is group.

Now, for Abelian we have to Satisfy the Commutative Law,

(V) **Commutative Law** :- $a \in Q_+$, $b \in Q_+$

$$a * b = \frac{ab}{2}$$

$$a * b = \frac{ba}{2}$$

$$a * b = b * a$$

Hence, $\{Q_+ , *\}$ is Abelian group.

Que :- Show that set of N of all natural No. $\{1, 2, 3, 4, \ldots\}$ is an Abelian group w.r.t Multiplication

DIY

Que :- Show that the set of +ve Rational No. is Abelian under Composition $a * b = \frac{ab}{3}$.

DIY

6. Show that the set of positive real numbers forms a group with multiplication.

7. Is the set of all odd integers with addition of integers form a group ?

8. Show that the set of natural numbers N with addition is a semi-group.

9. In the following, find semi-group and group. If not then give reasons :

(i) The set of natural numbers N with respect to multiplication.

(ii) The set of even natural numbers with addition.

(iii) The set of even natural numbers with multiplication.

21. Show that the set Q_+ of all positive rational numbers is a group with respect to the composition '*' defined as follows :

$$a * b = \frac{1}{4} ab \text{ where } a, b \in Q_+ .$$

22. Prove that the set of real numbers **R** form a group for the binary operation 'o', where

$$a \circ b = a + b - 1 \quad \forall\, a, b \in \mathbf{R}.$$

23. Write short notes on : groups and semi-groups. [R.G.P.V. 2003]

24. Let $(A, *)$ be a group, show that $(A, *)$ is an abelian group if and only if

$$a^3 * b^3 = (a * b)^3 \quad \forall\, a, b \in A.$$ [R.G.P.V. June 2007]

Some Examples on Finite Group :-

Ex:1 Show that the set of cube root of unity is an Abelian group w.r.t multiplication.

Sol:- Cube Root of unity are obtained by solving the following Eqn :-

$$x^3 - 1 = 0 \Rightarrow x^3 = 1$$

$$(x-1)(x^2+x+1) = 0 \qquad \{\because (a^3-b^3) = (a-b)\times(a^2+b^2+ab)\}$$

$$x - 1 = 0, \quad x^2 + x + 1 = 0$$

$$\boxed{x = 1}, \quad a = 1,\ b = 1,\ c = 1$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$$

$$x = \frac{-1+\sqrt{3}i}{2}, \quad x = \frac{-1-\sqrt{3}i}{2}$$

Now, Let

$$\frac{-1+\sqrt{3}i}{2} = w$$

$$\frac{-1-\sqrt{3}i}{2} = w^2$$

$$1 = w^3 \quad \{\text{from Question } x^3 = 1\}$$

So, $G = \{1, w, w^2\}$

Now, we form a Composition Table :-

$\cdot$	1	w	w^2
1	1	w	w^2
w	w	w^2	1
w^2	w^2	1	w

Here, $w \cdot w^2 = w^3 = 1$

$w^2 \cdot w^2 = w^4 = w^3 \cdot w$

$= 1 \cdot w$

$= \underline{\underline{w}}$

① **Closer Property** :- $a \in G, b \in G \Rightarrow a \cdot b \in G \ \forall a, b \in G$

$$\omega \in G, \ \omega^2 \in G \Rightarrow \omega \cdot \omega^2 \in G$$
$$\omega^3 \in G$$
$$1 \in G.$$
$$\forall \ \omega, \omega^2 \in G$$

② **Associative Law** :- $(a \cdot b) \cdot c = a \cdot (b \cdot c) \ \forall a, b, c \in G.$

$$(1 \cdot \omega) \cdot \omega^2 = 1(\omega \cdot \omega^2)$$
$$\omega \cdot \omega^2 = \omega \cdot \omega^2$$
$$\omega^3 = \omega^3$$
$$1 = 1$$
$$1 \in G$$
$$LHS = RHS.$$

③ **Existence of Identity** :- $a \cdot e = a = e \cdot a \ \forall a \in G.$

$e = 1$ for multiplication

$$e = 1 \in G$$

④ **Existence of Inverse** :- $a \in G \ \exists \ a^{-1} \in G \Rightarrow$

$$a \cdot a^{-1} = e = a^{-1} \cdot a$$
$$1 \in G \ \exists \ (1)^{-1} \in G \Rightarrow$$
$$1 \cdot (1)^{-1} = 1$$
$$1 \in G.$$

⑤ **Commutative Law** :- $a \in G, b \in G \Rightarrow a \cdot b = b \cdot a$

$$\omega \in G, \ \omega^2 \in G \Rightarrow \omega \cdot \omega^2 = \omega^2 \cdot \omega$$
$$\omega^3 = \omega^3$$
$$1 = 1$$
$$\therefore 1 \in G$$

Hence, $(G, \cdot)$ is Abelian group.

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Que :- Show that the set of four fouth root of unity (namely :- 1, -1, i, -i) forms an abelian group w.r.t multiplication.

Sol :- Let, $x^4 = 1 \Rightarrow x = 1^{1/4}$

$$x^4 - 1 = 0$$

$$(x^2 + 1)(x^2 - 1) = 0$$

$$x = 1, -1, i, -i$$

$$\therefore G = \{1, -1, i, -i\}.$$

To show that multiplication is a composition in G. we form a composition table as under :-

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

$\{\because i^2 = -1\}$

① closure Property :-

$\because$ all the entries in the composition table are the elements of the set G.

$\therefore$ the set G is closed w.r.t the multiplication.

② Associative Law :- $a, b, c \in G \Rightarrow (a \cdot b) \cdot c = a \cdot (b \cdot c)$ $\forall a, b, c \in G$

$1, -1, i \in G \Rightarrow (1 \cdot (-1)) \cdot i = 1((-1) \cdot i)$

$$-i = -i$$

$$-i \in G$$

$\forall 1, -1, i \in G$

③ **Existence of Identity** :- $a \cdot e = a = e \cdot a$

$e = 1$ for multiplication

$1 \in G$

④ **Existence of Inverse** :- $a \in G \; \exists \; a^{-1} \in G \Rightarrow$

$a \cdot a^{-1} = e = a \cdot a^{-1}$

$1 \in G \; \exists \; (1)^{-1} \in G$

$1 \cdot (1)^{-1} = 1 \in G$

⑤ **Commutative Law** :- $a \in G, \; b \in G \Rightarrow a \cdot b = b \cdot a \in G$

$\forall a, b \in G$

$1 \in G, \; i \in G \Rightarrow$

$1 \cdot i = i \cdot 1$

$i = i$

$i \in G$

Hence, $(G, \cdot)$ is Abelian group

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